Masking-Based Cross-Modal Remote Sensing Image–Text Retrieval via Dynamic Contrastive Learning

Zuopeng Zhao, Xiaoran Miao, Chen He, Jianfeng Hu, Bingbing Min, Yumeng Gao, Ying Liu, and Kanyaphakphachsorn Pharksuwan

***Abstract*— Cross-modal remote sensing image–text retrieval (CMRSITR) aims to extract comprehensive information from diverse modalities. The primary challenge in this field is developing effective mappings between visual and textual modalities to a shared latent space. Existing approaches generally focus on utilizing pretrained unimodal models to independently extract features from each modality. However, these techniques often fall short of achieving the critical alignment necessary for effective cross-modal matching. These techniques predominantly concentrate on the extraction of features and alignment at an instance level, suggesting potential areas for enhancement. To address these limitations, we introduce the masked interaction inferring and aligning (MIIA) framework, utilizing dynamic contrastive learning (DCL). This framework is adept at discerning the intricate relationships between local visual–textual tokens, thereby significantly bolstering the congruence of global image–text pairings without relying on additional prior supervision. Initially, we devise a masked interaction inferring (MII) module, which fosters token-level interplays through a novel masked visual-language (VL) modeling approach. Following this, we implement a cross-modal DCL mechanism, which is instrumental in capturing and aligning semantic correlations between images and texts more effectively. Finally, to ensure the comprehensive matching of visual and textual embeddings, we introduce a unique technique known as bidirectional distribution matching (BDM). This method is designed to minimize the Kullback–Leibler (KL) divergence between the distributions of image–text similarity, computed using the negative queues in momentum contrast learning. Comprehensive experiments performed on well-established public datasets consistently validate the state-of-the-art performance of MIIA methods in the CMRSITR task.**

***Index Terms*— Cross-modal remote-sensing image–text retrieval (CMRSITR), masked image modeling (MIM), masked language modeling (MLM), momentum contrast.**

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I. Introduction

IN RECENT decades, remote sensing (RS) technology has undergone rapid development, significantly enhancing our ability to monitor and understand Earth's environment [1], [2]. This abundance of RS data has opened unprecedented opportunities for a variety of applications, particularly in key areas such as agricultural monitoring, urban planning, environmental conservation, and response to natural disasters [3], [4], [5], [6], [7]. However, alongside these opportunities exists a significant challenge: effectively extracting useful information from the vast amounts of RS data and converting it into actionable insights [8], [9], [10]. This challenge has spurred an urgent need for more advanced RS image processing and analysis techniques [11], [12], including cross-modal RS image–text retrieval (CMRSITR) [13].

However, considering the significant variability in data distribution within RS scenes and the inherent heterogeneity between visual and linguistic modalities, the task of CMRSITR remains a formidable challenge. The first aspect of this challenge stems from the unique high spatial and spectral resolutions characteristic of RS images, further complicated by diversity in perspectives, scales, temporal–spatial changes, and environmental conditions. On the textual description front, challenges arise from the variability in the order of description and the high degree of similarity. Another significant challenge in tasks that bridge modalities, notably in cross-modal contexts, arises from inherent discrepancies in how vision and language are represented. To address this challenge, the main research objective in CMRSITR is to discover more efficient strategies for extracting unique feature representations. In addition, the research aims to enhance methods for matching different modalities and aligning images and texts in a unified embedding space. This approach is essential to ensure accurate and seamless integration of visual and textual elements in RS applications.

In the realm of natural language processing (NLP), the paradigm of pretraining followed by fine-tuning has proven to be highly effective. This methodology entails initially pretraining models on extensive datasets, exemplified by BERT [14], RoBERTa [15], and GPT-3 [16], followed by precise fine-tuning for designated downstream tasks. Recent developments have extended the utility of this paradigm to the

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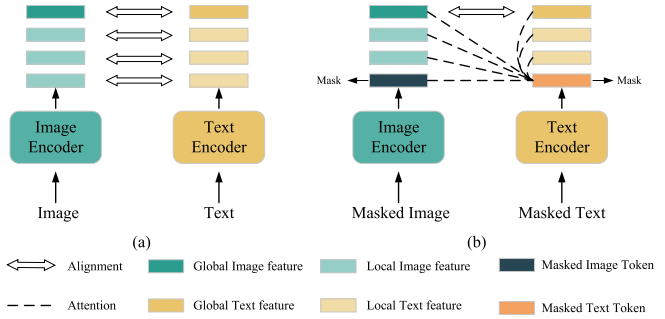


Fig. 1. Evolution of the CMRSITR method. (a) Global–local matching methods, which extract and align global–local image and text embeddings. (b) Our MIIA is an approach that infers the relationships between all local tokens and achieves better alignment of global image and text embeddings.

domain of visual-language (VL), as evidenced in recent studies [17], [18], [19]. The adoption of VL pretraining (VLP) has resulted in notable performance improvements across a variety of VL tasks, such as image–text retrieval, video–text retrieval, and visual question answering. Currently, VLP models represent a core research paradigm. Nonetheless, the exploration of efficient knowledge transfer from image–text pairs in natural domains to the more intricate realm of RS remains at an early stage.

First, the predominant approach in CMRSITR currently employs a single-stage method based on embeddings. This technique aims to synchronize images and texts within a unified embedding space by formulating a cross-modal matching loss function. However, certain implementations of this approach do not adequately facilitate modal interaction within intermediate network layers, a critical factor for bridging the modal disparity. While some strategies seek to augment retrieval efficiency by linking local features with textual elements, this may inadvertently introduce noise and uncertainty, as depicted in Fig. 1(a). In addition, the extraction and storage of numerous local features for both images and texts significantly increase the computational resource demands. This increase poses a notable challenge, thereby limiting the practical applicability of such methodologies in large-scale application scenarios.

To overcome these obstacles, we introduce the masked interaction inferring and aligning (MIIA) framework, as illustrated in Fig. 1(b). The MIIA framework skillfully establishes linkages between visual and textual representations through the utilization of a combination of self-attention and cross-attention mechanisms. MIIA leverages these amalgamated representations to conduct masked vision–language modeling (MVLM) tasks, thereby synergistically promoting intricate intra and intermodal relational learning. Although MVLM is predominantly utilized in the pretraining phase of VLP [20], [21], [22], [23], our research distinctively explores the applicability of MVLM in downstream fine-tuning tasks. The key innovation of our work lies in the development of a multimodal interaction encoder designed to adeptly amalgamate visual and textual representations. Utilizing the MVLM task, this encoder effectively aligns cross-modal fine-grained features, thereby allowing the backbone network to derive more distinctive global image–text representations. This process is achieved without requiring additional supervisory input.

Second, traditional multimodal tasks are often trained on limited datasets. To enhance the model's adaptability to varying modal data distributions and improve its generalization capabilities, we incorporate a dynamic dictionary mechanism combined with momentum contrast learning and negative sample sampling. Within this framework, we establish two sample queues that capture more intricate and nuanced cross-modal relationships through dictionaries that are dynamically updated. Furthermore, we have developed a unique KL alignment learning objective that bridges the gap between text-to-image and image-to-text retrieval tasks. For each image–text pair, we utilize the negative image queue generated in the momentum contrast learning process. By selecting texts from this queue and evaluating their similarity with noncorresponding images, we generate a probability distribution for the text-to-image retrieval task. A parallel process is employed to formulate a similar probability distribution for the image-to-text retrieval task. This methodology enhances not only the alignment between different modalities but also enriches the learning experience by integrating a more diverse array of sample interactions. This approach allows for a more robust and refined multimodal learning framework, capable of handling complex data distributions with improved efficiency.

Finally, to address the constraints of separate pretraining models in multimodal datasets, we adopt the contrastive language–image pretraining (CLIP) [19] as the foundational initialization for our model. CLIP demonstrates robust inherent proficiency in cross-modal alignment, which serves as a valuable asset for our objectives. By integrating our proposed MIIA Network, we effectively harness and transfer the extensive knowledge from the fully pretrained CLIP model. This enables us to further refine and learn detailed cross-modal relationships specifically on an image–text RS retrieval dataset. Moreover, in comparison to several contemporary methodologies [21], [24], [25], our MIIA framework exhibits greater efficiency, particularly during the inference phase where it calculates a singular global image–text similarity score. This streamlined approach significantly enhances the practicality and applicability of our model in a variety of large-scale RS scenarios. Through simplifying the computational process while maintaining accuracy, MIIA stands as a promising advancement in the field of multimodal data processing and retrieval.

The contributions of our work can be summarized as follows.

1) We propose a novel and efficient MIIA framework, adept at leveraging cross-modal fine-grained features to augment global alignment. This approach achieves its objectives without requiring extra supervisory input or additional inference costs.
2) We develop a cross-modal momentum contrast learning mechanism DCL, and introduce a novel cross-modal matching loss BDM. Experiments demonstrate that the DCL and the BDM loss based on it can effectively enhance performance.
3) Our results indicate that a CLIP model, when directly fine-tuned, is comparable to contemporary advanced methods. Furthermore, integrating our MIIA module

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facilitates the learning of nuanced image–text relationships, enhancing MIIA's ability to discern more distinctive image–text representations.

4) Extensive experiments on three public benchmark datasets, i.e., RSICD [26], RSITMD [27], and UCM [28] show that MIIA consistently outperforms the state-of-the-arts by a large margin.

This article is structured as follows. In Section II, we provide a brief overview of the related work in the fields of CMRSITR and VLP models. In Section III, we delve into the details and underlying principles of our proposed methodology. In Section IV, we conduct extensive comparative and ablation experiments to validate the effectiveness of our approach. Finally, we conclude our work in Section V, summarizing our findings and contributions.

II. Related Work

A. Cross-Modal RS Image–Text Retrieval

The task of CMRSITR is to retrieve corresponding RS images using the semantic information in descriptive texts. Due to the heterogeneity of RS image data and the semantic disparity between texts and images, the task of CMRSITR is extremely challenging. In addition, the wide distribution of RS images and the high similarity of texts pose certain limitations to the current work in this area. From an implementation perspective, the task of CMRSITR can generally be divided into title-based and embedding-based methods.

Caption-based retrieval methods in the domain of CMRSITR are typically categorized as two-stage retrieval approaches. These methods initially employ a caption generator to label each RS image within the database. Subsequently, during the retrieval phase, the similarity between the query text and the generated captions is calculated using various evaluation metrics [29], [30]. Qu et al. [28] pioneered the use of multimodal deep networks for semantic interpretation of high-resolution RS images. To enhance the representation of RS images, Shi and Zou [31] developed an RS image captioning (RSIC) framework, leveraging deep learning and a fully convolutional network to segment ground elements at diverse semantic scales. Lu et al. [26] introduced a substantial RSIC dataset, whereas Sumbul et al. [32] proposed a summary-driven RSIC method, aimed at addressing the challenge of information scarcity, and demonstrated its efficacy across various RS text–image datasets. In addition, Li et al. introduced an innovative truncated cross-entropy loss to reduce overfitting in RSIC [33]. Furthermore, Zhang et al. [34] combined global and local visual features, eliminating redundant elements to highlight more significant characteristics. Finally, Wang et al. [35] implemented multilabel scene categorization to provide comprehensive and precise semantic details. Despite the relative maturity of two-stage CMRSITR methods based on captions, they incur additional training costs and potentially magnify errors.

Embedding-based single-stage methods in CMRSITR primarily focus on directly measuring similarity by aligning data from disparate modalities into a unified subspace. Abdullah et al. [36] introduced an innovative CMRSITR framework utilizing a deep bidirectional triplet network. Furthermore, Yuan et al. [27] explored fine-grained multiscale approaches for text–image retrieval, presenting an asymmetric multiscale feature matching network. They also formulated a streamlined and potent cross-modal retrieval model tailored for multiscale and target-redundant RS data [37]. Moreover, acknowledging the significance of local information in depicting object relationships and salience, Yuan et al. [38] devised a dynamic fusion module for the integration of global and local visual features. Subsequently, Rahhal et al. [39] expanded the scope by incorporating multilanguage textual features, including languages beyond English, and developed a multilanguage framework based on transformers. In addition, some researchers have explored the establishment of connections between RS images and texts using graph neural networks [40]. Despite these advancements in embedding-based single-stage methods, they predominantly concentrated on enhancing the alignment of cross-modal features, often neglecting the unique characteristics of RS data distribution and the homogeneity of feature distribution within high-dimensional common spaces. These methods also overlooked the varying complexities of different sample pairs. The recent introduction of the multilanguage transformer [39] demonstrated that utilizing the CLIP pretrained model [19] could yield substantial results in CMRSITR tasks.

B. Vision–Language Pretraining Model

VLP focuses on comprehending the semantic correlations between visual and linguistic elements through pretraining on extensive image–text pair datasets. Inspired by the successes of transformer-based language model pretraining (e.g., BERT [14], XLNet [41], and BART [42]), and the visual transformer (ViT) [43], VLP has emerged as the leading approach for acquiring multimodal representations. This method has achieved noteworthy accomplishments in downstream tasks like image captioning [44], image–text retrieval [45], and visual question answering [46]. Current VLP research is divided into single-stream and two-stream models based on their structural design. Single-stream models [18], [23] merge text and visual features before processing them through a transformer encoder. This configuration is more efficient in terms of parameters as it utilizes a single parameter set for both modalities. However, its retrieval speed during inference is slower due to the necessity of computing similarity scores for all possible image–text combinations. Conversely, two-stream models [19], [47], [48] employ two distinct encoders for extracting text and visual features separately, without shared parameters. Although these models excel in image–text retrieval tasks, they are less capable of modeling intricate interactions between vision and language in other VL understanding tasks.

The use of pretrained models for RS image research has also undergone significant advancements [49]. Zhang et al. [50] introduced a transfer learning method from natural scenes to the RS domain, yielding impressive results in scene classification, object detection, and land cover classification. Nonetheless, the absence of a large and cohesive RS image–text dataset currently hinders the application of VLP

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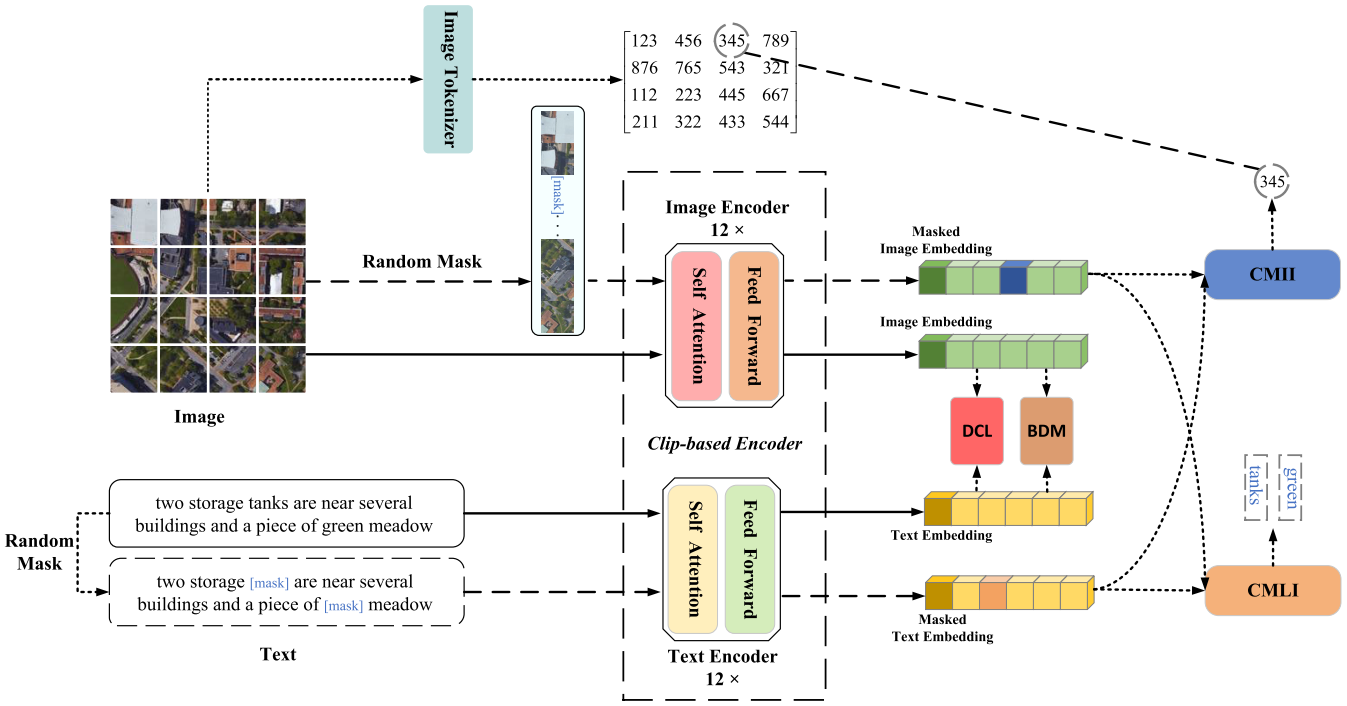


Fig. 2. Overview of the proposed MIIA framework. It comprises a dual-stream feature extraction backbone and three representation learning branches, i.e., masked interaction inferring (MII), dynamic contrastive learning (DCL), and bidirectional distribution matching (BDM). It utilizes these three tasks for end-to-end training. During the inference phase, the modules depicted with dashed lines are disabled, thus enhancing computational efficiency by solely calculating a single global image–text similarity score.

models in the RS domain. Liu et al. [21] proposed the first foundational model for RS in the realm of VL. This model is designed to learn robust visual features with rich semantics and aligned text embeddings, aiming to facilitate seamless downstream applications such as zero-shot image classification, linear probing, and k-NN classification, among others. Yuan et al. [24] explored parameter-efficient transfer learning (PETL) methods based on pretrained CLIP models. Their research focuses on effectively and efficiently transferring VL knowledge from natural domains to the CMRSITR tasks.

III. Method

This section will introduce the MIIA framework we proposed. Fig. 2 presents an overview of MIIA, with specific details to be discussed in Sections III-A–III-D.

A. Formulation

In the domain of cross-modal retrieval, it becomes imperative to compute the similarity metrics between RS imagery and corresponding text. Traditional methodologies typically involve encoding the image (denoted as I) and text (denoted as T) as distinct vectors, which are subsequently projected into a unified representational space. Drawing on the partial successes observed in the application of the CLIP model for CMRSITR, our approach involves directly initializing MIIA with the comprehensive CLIP image and text encoders. This strategy is aimed at significantly bolstering the inherent capabilities of MIIA in achieving fundamental cross-modal alignment.

Consider a cross-modal RS dataset denoted as $N = \{(I_n, T_n)\}_{n=1}^{N}$, comprising n positive pairs of image–text instances (I_n, T_n). For the sake of notational simplicity, we represent each RS image as $I \in R^{H\times W\times 3}$ and the corresponding text as $T = \{t_l\}_{l=1}^{L}$, where L is the length of the sentence. Here, $H \times W \times 3$ denotes the dimensions of the RS images, and t_l signifies the lth word in the text sequence.

In our methodology, we initiate by generating fixed-size, nonoverlapping patches denoted as $I_{\text{patch}} \in R^{(H/p)\times(W/p)\times D_v}$, utilizing a pretrained discrete variational autoencoder (dVAE) [51]. Here, p represents both the stride of the underlying backbone network and the dimensions of each patch, while D_v signifies the hidden dimension within the ViT. To facilitate classification, an additional token, $I_{\text{cls}} \in R^{1\times D_v}$, is appended to the sequence of tokens. Moreover, the ViT integrates positional embeddings $I_{\text{pos}} \in R^{N_v\times D_v}$ into each token, where $N_v = 1 + (H/p) \times (H/p)$, accounting for both the image patches and the classification token. The formulation of RS image tokens can be summarized as follows:

$$I_c = [I_{\text{cls}}; I_{\text{patch}}] + I_{\text{pos}} \tag{1}$$

where $I_c \in R^{N_v\times D_v}$ represents the composite token sequence. Following the integration of positional embeddings and the addition of the classification token, this enhanced token sequence is input into an N-layer transformer module. This module capitalizes on the relevance of each patch via a mechanism of masked self-attention. Subsequently, the [CLS] token is extracted and employed as the global RS image embedding, denoted as v. This embedding is then linearly projected into a D-dimensional cross-modal semantic space

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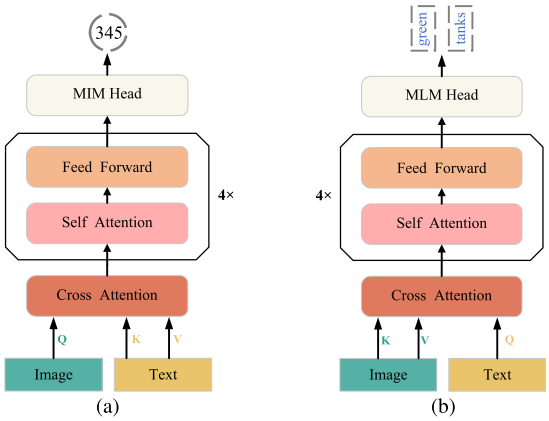


Fig. 3. Illustration of our MII, i.e., (a) CMII and (b) CMLI. (Masked) Visual and (masked) textual features are initially fused into a cross-attention layer, subsequently processed through transformer layers, and finally pass through either a MIM or masked language modeling (MLM) head.

and subsequently subjected to $\mathcal{L}_2$ normalization to ensure uniformity in its representation.

The text encoding process parallels the aforementioned image encoding, commencing with the tokenization of captions. This tokenization is executed using byte pair encoding (BPE), applied in lowercase, yielding a tokenized representation $T_{\text{token}} \in R^{N_t \times D_t}$. Each caption's token sequence initiates with a [SOS] token and culminates with a [EOS] token. Subsequently, these tokens are each augmented with positional embeddings, denoted as $T_{\text{pos}} \in R^{N_t \times D_t}$, to encode the sequential nature of the text. The formulation of text tokens can be summarized as follows:

$$T_c = T_{\text{pos}} + T_{\text{token}} \tag{2}$$

where $T_c \in R^{N_t \times D_t}$ encapsulates the finalized token sequence, with D_t signifying the hidden dimension of the text transformer. Postword embedding, these tokens are directed to an N-layer stacked transformer module. This module facilitates the assessment of each token's relevance through a mechanism of masked self-attention. The process culminates in the extraction of the global caption embedding, denoted as t, from the topmost layer of the transformer at the [EOS] token. This global embedding is then linearly projected into a D-dimensional cross-modal semantic space. Subsequently, it undergoes $\mathcal{L}_2$ normalization to ensure standardized representation across the semantic space.

B. Masked Interaction Inferring

To effectively harness fine-grained information and bridge the substantial modal gap between visual and linguistic data, a meticulous approach is paramount. While most existing methodologies focus on directly aligning features between images and text, our strategy, drawing inspiration from BERT [14] and BEIT [52], introduces a novel approach. As delineated in Fig. 3, our method is specifically engineered around an MII module, designed to enhance token-level interactions. This MII bifurcates into two distinct components: cross-attention masked image inferring (CMII) and cross-attention masked language inferring (CMLI).

Taking CMII as an exemplar, it encompasses a multihead cross-attention (MCA) layer, a four-layer transformer, and a masked image modeling (MIM) head. For each original image I, we initially utilize a dVAE to generate a $p \times p$ sequence of discrete image tokens I_{patch}. These tokens are then randomly masked with a probability of 40%, substituting the selected tokens with a special token [MASK]. Concurrently, the resultant masked image, denoted as I_m, is fed into the ViT as elaborated in Section III-A. The final hidden states of the visual and text transformers, $\{f_n^{i_m}\}_{n=1}^N$ and $\{f_n^t\}_{n=1}^L$, respectively, are then jointly input into the CMII process. To facilitate more effective fusion of masked image and text representations, the masked image representation $\{f_n^{i_m}\}_{n=1}^N$ is employed as the query ($\mathcal{Q}$), while the text representation $\{f_n^t\}_{n=1}^L$ serves as both the key ($\mathcal{K}$) and value ($\mathcal{V}$). This approach enables comprehensive interactions between the image and masked text representations, which can be articulated as follows:

$$\left\{f_n^{ic}\right\}_{n=1}^N = \text{XFMR}(\text{MCA}(\text{LN}(\mathcal{Q}, \mathcal{K}, \mathcal{V}))) \tag{3}$$

where $\{f_n^{ic}\}_{n=1}^N$ is the contextual representation that fuses the masked image and text, N is the length of the input image tokens, XFMR$(\cdot)$ is the transformer layer, LN$(\cdot)$ is layer normalization, and MCA$(\cdot)$ is multihead cross-attention, which can be implemented as follows:

$$\text{MCA}(\mathcal{Q}, \mathcal{K}, \mathcal{V}) = \text{softmax}\left(\frac{\mathcal{Q}\mathcal{K}^{\text{T}}}{\sqrt{d}}\right)\mathcal{V} \tag{4}$$

where d is the embedding dimension of the masked tokens.

For each masked position $\{f_n^{ic} : n \in M_i\}_{n=1}^N$, we use an MLP classifier to predict the probability of the corresponding original token

$$\left\{p_q^n\right\}_{q=1}^{|V_i|} = \text{MLP}\left(f_n^{ic}\right) \tag{5}$$

where M_i represents the set of masked image tokens, V_i is the size of the image token vocabulary, and p_q is the predicted token probability distribution. The loss for MII is as follows:

$$\mathcal{L}_{\text{cmii}} = -\frac{1}{|M_i||V_i|}\sum_{a\in M_i}\sum_{q\in|V_i|} y_q^n \log\frac{\exp\left(p_q^n\right)}{\sum_{k=1}^{|V_i|}\exp\left(p_k^n\right)} \tag{6}$$

where y_q is the one-hot encoded vocabulary distribution, in which the ground-truth token is assigned a probability of 1.

In a parallel vein, for each text T, the objective of CMLI is to accurately predict the label of each masked word token. This prediction is contingent upon the surrounding unmasked word tokens and the contextual information derived from the paired image. By utilizing the unmasked tokens and the image as references, CMLI seeks to enhance the model's understanding of the intricate interplay between textual and visual modalities, thereby facilitating a more robust and coherent integration of the two domains. The process for CMLI can be summarized as follows:

$$\mathcal{L}_{\text{cmli}} = -\frac{1}{|M_t||V_t|}\sum_{a\in M_t}\sum_{q\in|V_t|} y_q^n \log\frac{\exp\left(p_q^n\right)}{\sum_{k=1}^{|V_t|}\exp\left(p_k^n\right)} \tag{7}$$

where M_t represents the set of masked image tokens, V_t is the size of the image token vocabulary, and p_q is the probability of the corresponding original text tokens.

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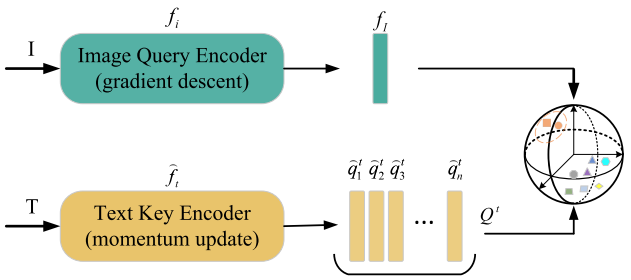


Fig. 4. Illustration of DCL. Two memory banks are established to store representations of the two modalities. The query encoder is updated using gradient descent, while the key encoder is updated through a momentum-based approach. This is an example of image-to-text retrieval, text-to-image retrieval follows a similar methodology.

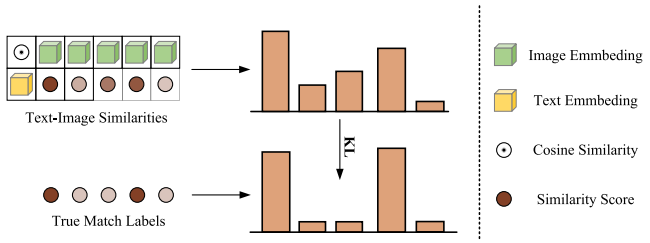


Fig. 5. Illustration of BDM. BDM aims to minimize the Kullback–Leibler (KL) divergence between the distributions of image–text similarity scores and the distributions of true label matchings. This approach not only effectively increases the variance among nonmatching pairs but also simultaneously enhances the correlation among matching pairs.

Subsequently, the total loss, which is a composite of the losses from both CMII and CMLI, is calculated. This total loss serves as a comprehensive measure of the model's performance in bridging the gap between the visual and textual modalities, enhancing its ability to process and interpret cross-modal data. The total loss function can be articulated as follows:

$$\mathcal{L}_{\text{mii}} = \mathcal{L}_{\text{cmii}} + \mathcal{L}_{\text{cmli}}. \tag{8}$$

C. Dynamic Contrastive Learning

Research indicates that utilizing a large number of negative samples can significantly enhance retrieval performance in image–text retrieval tasks [19]. However, traditional batch contrastive learning methods are often constrained by the small batch size due to GPU memory limitations, which restricts the number of negative samples. To overcome this limitation, we propose a cross-modal DCL algorithm, based on MoCo [53], aimed at enhancing instance-level interactions between images and texts. Unlike the MoCo algorithm, which supports learning in a single modality through one feature queue (typically images), our DCL algorithm maintains two separate feature queues and promotes mutual learning between different modalities through enhanced cross-modal feature interactions. This approach effectively decouples the number of negative samples from the batch size, thereby enhancing computational feasibility and efficiency.

As shown in Fig. 4, we define f^i (with parameters θ^i) and f^t (with parameters θ^t) as the encoders for the image and text modalities, respectively. To complement these, we employ two additional momentum encoders: $\hat{f}^i$ (with parameters $\hat{\theta}^i$) for the visual modality, and $\hat{f}^t$ (with parameters $\hat{\theta}^t$) for the language modality. The parameters of these momentum encoders are updated in a manner that ensures gradual and consistent learning, reflecting the most recent data representations while maintaining stability over time. The update rule for the parameters of the momentum encoders is as follows:

$$\hat{\theta}^i = \varsigma \cdot \hat{\theta}^i + (1-\varsigma)\cdot \theta^i \tag{9}$$

$$\hat{\theta}^t = \varsigma \cdot \hat{\theta}^t + (1-\varsigma)\cdot \theta^t \tag{10}$$

where ς is the momentum hyperparameter.

In addition, in our model, we maintain two distinct queues: $Q^i = \{\hat{q}_a^i\}_{a=1}^{N_q}$ for images and $Q^t = \{\hat{q}_a^t\}_{a=1}^{N_q}$ for texts. Here, $\hat{q}_a^i/\hat{q}_a^t$ represents the momentum feature vectors, while N_q denotes the size of each queue. For each mini-batch $\mathcal{B} = \{(I_n, T_n)\}_{n=1}^{N_b} \subseteq N(N_b = |\mathcal{B}| \ll N_q)$, the samples are processed through the current momentum encoders to obtain their respective momentum feature vectors. Postloss computation, these vectors are then appended to the corresponding queue.

Simultaneously, to maintain the queue size, the oldest N_b momentum feature vectors are removed from each queue. In this setting, for each image within a given data batch, its paired text is considered a positive sample, while all samples in Q^t are treated as negative samples. This arrangement facilitates the calculation of the contrastive loss from image to text, which can be formulated as follows:

$$\mathcal{L}_{I2T} = -\frac{1}{N_b}\sum_{(I,T)\in B}\log\frac{\exp(f_I\cdot\hat{f}_T/\tau)}{\sum_{\hat{q}_a^t\in Q^t}\exp\left(f_I\cdot\hat{q}_a^t/\tau\right)} \tag{11}$$

where $f_I = f^i(I)$, $\hat{f}_T = \hat{f}^t(T)$, and τ is the temperature hyperparameter.

In this approach, the similarity between two feature vectors is gauged using the dot product. Correspondingly, for each text in the data batch, we define the contrastive loss from text to image based on a similar methodology

$$\mathcal{L}_{T2I} = -\frac{1}{N_b}\sum_{(I,T)\in B}\log\frac{\exp(f_T\cdot\hat{f}_I/\tau)}{\sum_{\hat{q}_a^i\in Q^i}\exp\left(f_T\cdot\hat{q}_a^i/\tau\right)}. \tag{12}$$

The total loss is as follows:

$$\mathcal{L}_{\text{dcl}} = \mathcal{L}_{I2T} + \mathcal{L}_{T2I}. \tag{13}$$

D. Bidirectional Distribution Matching

We introduce a novel cross-modal matching loss function, termed BDM. This loss function aims to integrate the cosine similarity distribution of $N \times N$ text–image pair embeddings into the KL divergence. The primary objective of BDM is the effective correlation of representations from different modalities. The specific structure is shown in Fig. 5.

For each image–text pair $(I_n, T_n) \in \mathcal{B}$, we define the probability distribution for the image–text task as follows:

$$D_{I2T} = \left[p\left(f_I, \hat{f}_T\right), p\left(f_I, \hat{q}_1^t\right), \ldots, p\left(f_I, \hat{q}_{N_q}^t\right)\right] \tag{14}$$

where

$$p(f_I, \hat{f}_T) = \frac{\exp(f_I, \hat{f}_T/\tau)}{\sum_{K=1}^{N_q}\exp(f_I, \hat{f}_k/\tau)} \tag{15}$$

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TABLE I
OVERVIEW OF THREE MAINSTREAM CMRSITR DATASETS

	Image			Text Length	
Dataset	Number	Size	Class	Average	Maximum
RSICD [26]	10921	224 × 224	30	10.55	34
RSITMD [27]	4743	256 × 256	32	10.25	34
UCM [28]	2100	256 × 256	21	10.51	22

and $p(f_I, \hat{q}_j^t)(j = 1, 2, \ldots, N_q)$ can be calculated in the same manner. Similarly, the probability distribution for the text-to-image task can be obtained as

$$D_{T2I} = \left[p\left(f_T, \hat{f}_I\right), p\left(f_T, \hat{q}_1^i\right), \ldots, p\left(f_T, \hat{q}_{N_q}^i\right)\right]. \quad (16)$$

Therefore, our learning objective for BDM is formulated as: minimizing the symmetric KL divergence between D_{I2T} and D_{T2I}

$$\mathcal{L}_{\text{bdm}} = \frac{1}{N_b} \sum_{(I_n, T_n) \in B} \left(\text{KL}\left(D_{I_{2T}} \middle\| D_{T_{2I}}\right)\right) + \left(\text{KL}\left(D_{T_{2I}} \middle\| D_{I_{2T}}\right)\right). \quad (17)$$

IV. EXPERIMENTS AND DISCUSSION

A. Dataset and Evaluation Metrics

To assess the efficacy of our MIIA framework, three prominent RS image datasets are utilized: RSICD [26], RSITMD [27], and UCM [28]. The RSICD dataset is an extensive RS image caption dataset that comprises 10 921 images with varying pixel resolutions, standardized to a fixed size of 224 × 224 pixels. Each image in the RSICD dataset is accompanied by five descriptive sentences. The RSITMD dataset, an extension of RSICD, includes some images from the RSICD dataset and additional images sourced from Google Earth. The UCM dataset, based on the UC Merced Land Use dataset [54], encompasses 2100 RS images. Each image in UCM is accompanied by manually annotated descriptive texts. A notable aspect of these datasets, especially pertinent to CMRSITR, is the high degree of similarity in descriptions of RS images from the same category [26], presenting an added layer of complexity for effective retrieval. For detailed information on the three datasets, please refer to Table I.

To validate the effectiveness of MIIA, we choose two widely used evaluation metrics: $R@k$ (k = 1, 5, 10) and mR [55]. R@k measures the proportion of true values that appear in the top k retrieval results. A relation label of 1 indicates that the RS image and text are a positive sample pair; otherwise, they are considered a negative sample pair. Initially, one must obtain the k relation labels corresponding to the top k matching scores between the query and all retrieval entities. Subsequently, it involves identifying how many positive samples are among the top k retrieved samples. Therefore, $R@k$ can be defined as

$$R@k = \frac{1}{B} \sum_{i=1}^{B} \text{Score}(i, k) \quad (18)$$

where B means the volume of the query set, i means the ith query set, and Score(i, k) denotes whether the positive items corresponding to ith query appear in the top k retrieval samples or not. If the positive sample exists in the top k retrieval results, Score(i, k) = 1. Otherwise, Score(i, k) = 0. Furthermore, mR is the average of all R@ks

$$\text{mR} = \left(\underbrace{R@1+R@5+R@10}_{\text{Text retrieval}} + \underbrace{R@1+R@5+R@10}_{\text{Image retrieval}}\right) \Big/ 6 \quad (19)$$

where "text retrieval" pertains to text-to-image retrieval, and "image retrieval" pertains to image-to-text retrieval.

B. Implementation Details

The MIIA architecture encompasses a pretrained image encoder (i.e., CLIP-ViT-B/16) and a pretrained text encoder (i.e., CLIP text transformer). Each layer of the MIIA is configured with a hidden size of 512 and 8 attention heads. In our approach, 40% of image patches are randomly masked, and text word tokens are masked with a 15% probability. The training process incorporates data augmentation techniques such as random horizontal flipping, random cropping with padding, and random erasing. All input images are resized to dimensions of 256 × 256. The momentum hyperparameter ς is set to 0.99 and the temperature parameter τ is set to 0.02. The queue sizes (N_q) are set to 1200 for RSICD, 800 for RSITMD, and 400 for UCM. We train the model for 60 epochs using the Adam optimizer [56], with an initial learning rate of 1×10^{-5} and a cosine learning rate decay. A 5-epoch warm-up period was employed to linearly increase the learning rate from 1×10^{-6} to 1×10^{-5}. For the randomly initialized module, we set the initial learning rate at 5×10^{-5}. We conducted these experiments using two RTX3090 24 GB GPUs.

C. Comparison With State-of-the-Art Methods

In our experimental research, we conducted a comprehensive comparative analysis of the MIIA framework, contrasting it not only with traditional cross-modal retrieval methods but also with those based on VLP. Notably, traditional methods and fine-tuned models such as CLIP and BLIP are predominantly trained on single datasets. In contrast, RemoteCLIP was trained on the dataset proposed in its paper, which includes RET-3, DET-10, and SEG-4 [21], whereas RS-CapRet was trained on the Cap-4 dataset [65]. Our training was solely conducted on RET-3. This comparison is intended to provide a thorough understanding of MIIA's performance relative to existing methods.

1) Traditional Methods: We compared the proposed method with image–text retrieval models (VSE++ [57], SCAN [58], CAMP [59], AMFMN [27], GaLR [38], and HyperMatch [61]) on RS image–text datasets. For these methods, we used results from three publications [21], [24], [25]. In addition, we also included three of the latest RSITR methods.

1) *CABIR [60]:* CABIR employs an innovative cross-modal attention mechanism, wherein the model utilizes textual semantics to assign weights to specific regions in an image. This approach efficiently filters out superfluous

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features, thus reducing the influence of irrelevant scene semantics on the retrieval process.

2) *HVSA [62]:* HVSA introduces an innovative hypersphere-based visual-semantic alignment network through curriculum learning, treating sample pairs of different difficulties unequally. This approach leads to better embedding representations by projecting features onto a unit hypersphere while differentiating between samples based on their complexity.
3) *IEFT [25]:* IEFT introduces a simplified feature embedding module that is designed to map images and texts into distinct visual and textual feature spaces. Furthermore, it has incorporated an information interaction enhancement module, specifically designed to concurrently model the inherent relationships between RS images and texts, while also augmenting the visual features.

2) VLP-Based Methods: Whether it is full-tuning or adapter-based [66], [67] approaches, both are currently hot topics in research, with VLP-based Methods often yielding surprisingly effective results. We have compared recent VLP-based methods as follows.

1) *CLIP [19]:* CLIP is trained using large-scale image–text pairs with an image encoder and a text encoder to map images and text into a common vector space. Within this common vector space, CLIP learns cross-modal representations by maximizing the cosine similarity among relevant image–text pairs while minimizing it between unrelated pairs. This contrastive learning framework effectively captures and associates information across both modalities and endows the model with exceptional zero-shot recognition capabilities.
2) *BLIP [63]:* BLIP has been proposed for unified understanding and generation tasks. This framework incorporates an innovative dataset bootstrapping method, named CapFilt, which leverages noisy web image–text pairs for learning. BLIP employs a multimodal encoder–decoder hybrid system (MED), effectively handling various VL tasks such as image–text retrieval, image captioning, and visual question answering, thereby achieving state-of-the-art performance in these areas.
3) *X2-VLM [64]:* X2-VLM represents a versatile pretrained model designed for visual and linguistic tasks. It is capable of learning multigranularity alignments between images and text, which include object labels, region annotations, and comprehensive image descriptions. The model features a modular architecture that includes visual, textual, and fusion modules, all based on transformer structures.
4) *Remote-CLIP [21]:* The first-ever VL foundational model for RS is developed with the objective of learning potent visual features imbued with rich semantics, alongside aligned text embeddings. This design aims to facilitate smooth and efficient downstream applications in the field.
5) *PE-RSITR [24]:* PE-RSITR introduces a PETL approach capable of effectively and efficiently transferring VL knowledge from natural domains to the RS domain. By utilizing CLIP in image–text retrieval tasks, this method specifically addresses the challenges and impracticalities associated with comprehensive fine-tuning in real-world applications.
6) *RS-CapRet [65]:* This method enhances RS image description and text–image retrieval tasks by integrating a high-performance large decoder language model with an image encoder specifically adapted for RS imagery. By leveraging contrastive language-image pretraining, this combination effectively describes and retrieves RS images.

Analyzing the data in Tables II–IV, we note that methods based on VLP significantly surpass traditional approaches across all metrics. In the natural domain, text-to-image retrieval is generally considered more challenging than image-to-text retrieval, as reported in prior studies [63], [68]. However, Yuan et al. [69] conducted a quantitative analysis of text distributions in CMRSITR using BLEU and METEOR metrics, concluding that text similarity in the RS domain is significantly higher than in the natural domain. This observation undoubtedly increases the complexity of image-to-text retrieval tasks.

Taking the RSICD dataset as an example, the best performance among traditional methods is achieved by the IEFT method, with a text retrieval R@1 score of 8.78%. In contrast, the zero-shot CLIP and BLIP methods nearly reach this result. Within the fully-tuned CLIP methods, the R@1 score is 13.54%, representing an improvement of over 50% compared to IEFT. The disparity between traditional methods and VLP-based methods is even more pronounced on the R@5 and R@10 text retrieval metrics, demonstrating the efficiency of VLP-based Methods in processing cross-modal RS data. The VLP-based models in the general domain have also achieved commendable results in the CMRSITR task. By examining the feature extraction column (Image and Text Enc.), it becomes evident that transformer-based methods have taken a leading role in CMRSITR. This trend highlights the efficacy and growing dominance of transformer architectures in efficiently handling and interpreting the complexities inherent in multimodal data processing. Significantly, our method attains 19.51%, 39.05%, and 53.84% in R@1, R@5, and R@10 image retrieval metrics, respectively, and scores the highest in mR at 37.81%, surpassing all other methods. These outcomes highlight the exceptional performance of our method in image and text feature learning, as well as in the fusion of cross-modal features.

As depicted in Fig. 6, our analysis reveals that CLIP-based models, such as remote-CLIP and MIIA, exhibit a higher parameter count compared to conventional models like GALR and HyperMatch. This parameter increase is primarily due to the utilization of transformer-based encoders in these models, which typically have a greater number of parameters than architectures like CNNs and BiGRUs. Notably, our MIIA model ranks second in terms of parameter count, trailing only behind remote-CLIP (ViT-L/14). The relatively high parameter count in our model stems from the integration of a cross-modal attention mechanism within the MII module, contributing

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TABLE II
PERFORMANCE COMPARISONS WITH STATE-OF-THE-ART METHODS ON RSICD DATASET

Method	Tuned on	Image Enc.	Text Enc.	Text retrieval			Image retrieval			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
Traditional methods:										
VSE++ [57]	RSICD	ResNet-18	GRU	3.38	9.51	17.46	2.82	11.32	18.10	10.43
SCAN t2i [58]	RSICD	Faster R-CNN	BiGRRU	4.39	10.90	17.64	3.91	16.20	26.49	13.25
SCAN i2t [58]	RSICD	Faster R-CNN	BiGRRU	5.85	12.89	19.84	3.71	16.40	26.73	14.23
CAMP-triplet [59]	RSICD	Faster R-CNN	BiGRRU	5.12	12.89	21.12	4.15	15.23	27.81	14.39
AMFMN-fusion [27]	RSICD	ResNet-18	BiGRRU	5.39	15.08	23.40	4.90	18.28	31.44	16.42
GALR-MR [38]	RSICD	ResNet-18	BiGRRU	6.59	19.85	31.04	4.69	19.48	32.13	18.96
CABIR [60]	RSICD	ResNet-152	BERT+BiGRU	8.59	16.27	24.13	5.42	20.77	33.58	18.12
HyperMatch [61]	RSICD	ResNet-18	BiGRRU	7.14	20.04	31.02	6.08	20.37	33.82	19.75
HVSA [62]	RSICD	Faster R-CNN	BiGRRU	7.47	20.62	32.11	5.51	21.13	34.13	20.16
IEFT [25]	RSICD	ViT	BERT	8.78	28.47	43.88	8.38	28.17	44.16	26.97
VLP-based Methods:										
Zero-Shot CLIP [19]	——	ViT-B/32	Transformer	6.77	15.37	23.15	5.01	15.75	24.21	15.04
Zero-Shot BLIP [63]	——	ViT-B/32	Bert	7.03	16.11	23.11	5.73	15.31	25.81	15.52
Fine-tuning CLIP [19]	RSICD	ViT-B/32	Transformer	13.54	30.83	43.46	11.55	33.14	49.83	30.39
Fine-tuning BLIP [63]	RSICD	ViT-B/32	Bert	13.81	30.27	43.61	12.77	32.71	49.92	30.52
X2-VLM [64]	RSICD	BEiT2	Bert	13.37	33.21	45.33	12.43	32.93	50.21	31.25
PE-RSITR [24]	RSICD	ViT-B/32	Transformer	14.13	31.51	44.78	11.63	33.92	50.73	31.12
RemoteCLIP [21]	RemoteCLIP dataset	ResNet-50	Transformer	10.76	32.83	48.75	13.36	32.94	44.83	30.58
RemoteCLIP [21]	RemoteCLIP dataset	ViT-B/32	Transformer	13.71	37.11	54.25	17.02	37.97	51.51	35.26
RemoteCLIP [21]	RemoteCLIP dataset	ViT-L/14	Transformer	14.73	39.93	56.58	18.39	37.42	51.05	36.35
RS-CapRet [65]	Cap-4	ViT-L/14	Transformer	10.25	31.62	48.53	—	—	—	30.13
Ours	RET-3	ResNet-50	Transformer	11.05	30.17	49.32	13.21	34.74	45.60	30.68
Ours	RET-3	ViT-B/16	Transformer	**15.82**	**40.92**	**57.71**	**19.51**	**39.05**	**53.84**	**37.81**

TABLE III
PERFORMANCE COMPARISONS WITH STATE-OF-THE-ART METHODS ON RSITMD DATASET

Method	Tuned on	Image Enc.	Text Enc.	Text retrieval			Image retrieval			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
Traditional methods:										
VSE++ [57]	RSITMD	ResNet-18	GRU	10.38	27.65	39.60	7.79	24.87	38.67	24.83
SCAN t2i [58]	RSITMD	Faster R-CNN	BiGRRU	10.18	28.53	38.49	10.10	28.98	43.53	26.64
SCAN i2t [58]	RSITMD	Faster R-CNN	BiGRRU	11.06	25.88	39.38	9.82	29.38	42.12	26.28
CAMP-triplet [59]	RSITMD	Faster R-CNN	BiGRRU	11.73	26.99	38.05	8.27	27.79	44.34	26.20
AMFMN-fusion [27]	RSITMD	ResNet-18	BiGRRU	11.06	29.20	38.72	9.96	34.03	52,96	29.32
GALR-MR [38]	RSITMD	ResNet-18	BiGRRU	14.82	31.64	42.48	11.15	36.68	51.68	31.41
HyperMatch [61]	RSITMD	ResNet-18	BiGRRU	11.73	28.10	38.05	9.16	32.31	46.64	27.67
HVSA [62]	RSITMD	Faster R-CNN	BiGRRU	13.20	32.08	45.58	11.43	39.20	57.45	33.16
IEFT [25]	RSITMD	ViT	BERT	15.49	37.61	51.40	11.19	38.09	58.84	35.44
VLP-based Methods:										
Zero-Shot CLIP [19]	——	ViT-B/32	Transformer	9.29	26.33	37.39	7.79	23.67	38.89	23.89
Zero-Shot BLIP [63]	——	ViT-B/32	Bert	10.03	26.81	49.13	9.05	25.14	39.71	26.65
Fine-tuning CLIP [19]	RSITMD	ViT-B/32	Transformer	24.16	47.12	61.28	20.40	50.53	**68.54**	45.33
Fine-tuning BLIP [63]	RSITMD	ViT-B/32	Bert	25.31	47.01	63.14	21.33	52.10	68.48	46.23
X2-VLM [64]	RSITMD	BEiT2	Bert	24.83	48.06	62.76	22.81	52.81	67.64	46.49
PE-RSITR [24]	RSITMD	ViT-B/32	Transformer	23.67	44.07	60.36	20.10	50.63	67.97	44.47
RemoteCLIP [21]	RemoteCLIP dataset	ResNet-50	Transformer	19.29	51.55	70.58	23.67	47.57	64.60	46.21
RemoteCLIP [21]	RemoteCLIP dataset	ViT-B/32	Transformer	22.17	56.46	73.41	27.88	50.66	65.71	49.38
RemoteCLIP [21]	RemoteCLIP dataset	ViT-L/14	Transformer	23.76	**59.51**	74.73	**28.76**	52.43	63.94	50.52
Ours	RET-3	ResNet-50	Transformer	19.82	52.10	68.11	23.41	45.21	63.29	45.32
Ours	RET-3	ViT-B/16	Transformer	**25.22**	58.81	**75.13**	27.91	**53.11**	66.18	**51.06**

to the overall parameter increase in the CLIP framework. However, this increased parameter investment is justified, as evidenced by the superior performance of our method. Furthermore, the advantages of our approach become even more apparent in terms of computational efficiency, a topic that will be elaborated upon in Section IV-D6.

D. Ablation Study

To validate the efficacy of the MIIA framework, we conduct detailed ablation experiments. First, we systematically analyze the performance using different CLIP backbones. Subsequently, we carry out a comprehensive comparative analysis of different momentum parameters ς and the cross-attention mechanism in MII. Finally, we examine the effectiveness of each component within the MIIA framework. This comprehensive approach not only underscores the strengths of our framework but also provides insights into the contributions of individual components to the overall performance.

1) Vision Backbones: Fig. 7 presents the results of different CLIP image encoders on two datasets. The experimental

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TABLE IV
PERFORMANCE COMPARISONS WITH STATE-OF-THE-ART METHODS ON UCM DATASET

Method	Tuned on	Image Enc.	Text Enc.	Text retrieval			Image retrieval			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
Traditional methods:										
VSE++ [57]	UCM	ResNet-18	GRU	12.38	44.76	65.71	10.10	31.80	56.85	36.93
SCAN t2i [58]	UCM	Faster R-CNN	BiGRRU	14.29	45.71	67.62	12.76	50.38	77.24	44.67
SCAN i2t [58]	UCM	Faster R-CNN	BiGRRU	12.85	47.14	69.52	12.48	46.86	71.71	43.43
CAMP-triplet [59]	UCM	Faster R-CNN	BiGRRU	10.95	44.29	65.71	9.90	46.19	76.29	42.22
AMFMN-fusion [27]	UCM	ResNet-18	BiGRRU	16.67	45.71	68.57	12.86	53.24	79.43	46.08
CABIR [60]	UCM	RN152	BERT+BiGRU	15.17	45.71	72.85	12.67	54.19	89.23	48.30
VLP-based Methods:										
Zero-Shot CLIP [19]	——	ViT-B/32	Transformer	10.95	34.76	55.71	7.33	35.14	54.57	33.08
Zero-Shot BLIP [63]	——	ViT-B/32	Bert	10.83	35.03	57.21	8.91	33.21	55.73	33.49
Fine-tuning CLIP [19]	UCM	ViT-B/32	Transformer	17.14	55.24	79.52	13.90	56.95	91.81	52.43
Fine-tuning BLIP [63]	UCM	ViT-B/32	Bert	18.21	56.44	83.14	15.43	56.33	89.21	53.13
X2-VLM [64]	UCM	BEiT2	Bert	19.33	57.31	80.74	16.44	58.19	90.17	53.70
PE-RSITR [24]	UCM	ViT-B/32	Transformer	**22.71**	55.81	80.33	18.82	**62.84**	**93.72**	55.71
RemoteCLIP [21]	RemoteCLIP dataset	ResNet-50	Transformer	15.24	57.14	84.57	13.33	50.48	74.76	49.25
RemoteCLIP [21]	RemoteCLIP dataset	ViT-B/32	Transformer	18.67	61.52	94.29	20.48	59.85	83.33	56.36
RemoteCLIP [21]	RemoteCLIP dataset	ViT-L/14	Transformer	17.71	62.19	93.90	19.05	54.29	80.95	54.68
RS-CapRet [65]	Cap-4	ViT-L/14	Transformer	16.10	56.29	90.76	—	—	—	54.38
Ours	RET-3	ResNet-50	Transformer	15.13	60.21	80.97	15.69	51.07	75.21	49.71
Ours	RET-3	ViT-B/16	Transformer	18.61	**63.41**	**95.16**	**21.17**	58.36	83.97	**56.78**

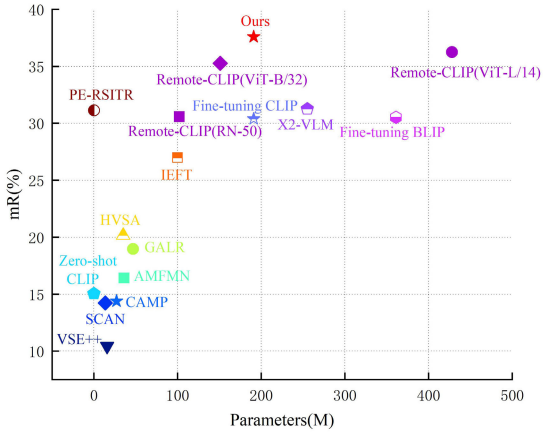


Fig. 6. Retrieval performance is correlated with the number of trainable parameters in the RSICD dataset.

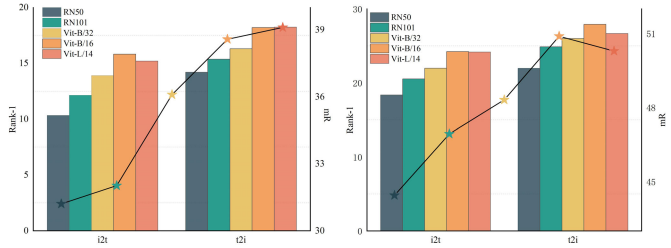


Fig. 7. Performance comparison of different image encoders on (a) RSICD and (b) RSITMD datasets.

outcomes indicate that more advanced networks yield better performance. However, the use of ViT-L/14 does not substantially outperform ViT-B/16. Given the parameter size of the model, we opted for ViT-B/16 as our image encoder. This choice strikes a balance between performance and computational efficiency, making ViT-B/16 a suitable choice for our framework.

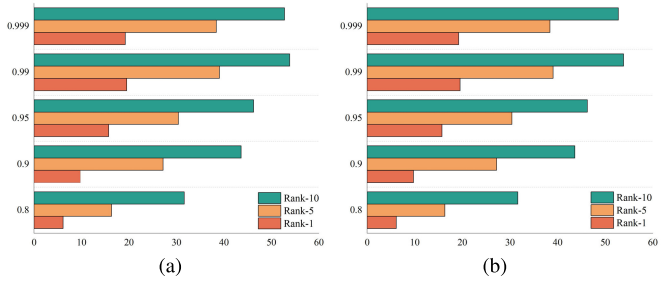


Fig. 8. Ablations of the momentum values on (a) RSICD and (b) RSITMD datasets.

2) Momentum Values ς: The momentum parameter ς is a crucial element in DCL. Experiments were conducted on the RSICD and RSITMD datasets using different momentum parameters, as shown in Fig. 8. The performance improves progressively when ς is increased from 0.99, indicating that a relatively larger momentum is beneficial. However, when ς is set to values lower than 0.9, there is a significant drop in accuracy. Therefore, we decided to set ς at 0.99, balancing the need for stability in the model's learning process with the requirement for effective information capture and update.

3) MII Module: To thoroughly assess the influence of various components within the MIIA, we undertake a detailed empirical analysis of two public datasets (i.e., RSICD and RSITMD). The performance metrics, specifically the accuracies for R@1, R@5, and R@10, are expressed as percentages and detailed in Table V. We apply CMII and CMLI tasks in our approach to elucidate local relationships, facilitating the seamless integration of MII with other transformer-based methods to enhance fine-grained cross-modal alignment. The efficacy of the MII component is validated through comparative experimental results, especially in comparisons between experiments No.0 and No.4, No.2 and No.6, as well as

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TABLE V
ABLATION STUDY ON EACH COMPONENT OF MIIA ON RSICD AND RSITMD

No.	Methods	Components			RSICD						RSITMD					
					Text retrieval			Image retrieval			Text retrieval			Image retrieval		
		BDM	DCL	MII	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
0	Baseline				16.18	35.82	51.01	12.78	36.02	53.23	25.11	46.91	64.67	20.98	57.16	71.33
1	$+\mathcal{L}_{cmpm}$				12.67	30.63	47.81	8.21	32.18	48.61	20.81	41.77	58.58	14.39	51.84	68.28
2	+BDM	✓			17.23	37.56	52.31	13.73	37.58	55.36	26.19	48.37	65.71	21.76	58.28	72.74
3	+DCL		✓		16.95	36.21	52.01	13.96	37.14	54.85	25.76	47.37	65.15	21.39	57.69	72.25
4	+MII			✓	17.93	38.54	52.86	14.74	38.19	55.71	26.31	49.21	65.24	22.17	57.35	72.77
5	+BDM+DCL	✓	✓		17.61	37.93	51.95	13.43	38.83	56.21	26.89	50.26	66.13	22.56	58.42	73.85
6	+BDM+MII	✓		✓	18.11	38.94	52.91	15.33	39.17	56.91	27.12	51.64	65.74	23.75	58.25	74.73
7	MIIA	✓	✓	✓	**18.21**	**39.05**	**53.84**	**15.82**	**40.92**	**57.71**	**27.91**	**53.11**	**66.18**	**24.23**	**58.81**	**75.13**

TABLE VI
COMPARISONS BETWEEN DIFFERENT MULTIMODAL INTERACTION MODULE OF MII ON RSICD

Method	Param(M)	Time(ms)	T2I rsum	I2T rsum	mR
Baseline	——	——	103.01	102.03	34.17
Self-attention	32.17	71.64	109.72	112.37	37.01
Merged attention	**13.66**	42.45	110.21	113.47	37.28
Ours	15.21	**20.30**	**111.10**	**114.45**	**37.59**

TABLE VII
COMPARISON OF ZERO-SHOT TRANSFER CAPABILITIES ACROSS DIFFERENT MODELS ON THE RSITMD AND UCM DATASETS

Method	RSITMD			UCM		
	T2IR@1	I2TR@1	mR	T2IR@1	I2TR@1	mR
CLIP	13.21	10.36	30.77	12.77	9.37	39.27
BLIP	12.33	9.94	29.60	11.52	10.52	41.56
X2-VLM	11.84	12.92	32.19	12.10	**12.37**	42.67
Ours	**14.82**	**12.61**	**34.64**	**13.18**	12.11	**43.25**

No.5 and No.7. The introduction of MII to the baseline model resulted in notable increases in R@1 accuracy on the RSICD and RSITMD datasets, improving by 1.75% and 1.20%, respectively. These findings clearly demonstrate the positive contribution of the MII module in enhancing cross-modal matching, thereby underscoring its pivotal role in the improvement of multimodal data correlation.

4) Dynamic Contrastive Learning: The experimental outcomes, particularly when comparing No.2 versus No.5 and No.6 versus No.7, clearly highlight the effectiveness of the momentum contrast learning mechanism within the DCL framework. Importantly, combining DCL with the MII component, as observed in No. 6, results in enhanced performance compared to the use of MII alone (as in No. 4). This finding indicates that DCL contributes significantly to the model's performance, even when integrated with advanced mechanisms like MII. These results emphasize the importance of the momentum contrast learning mechanism in augmenting the learning process. They confirm its pivotal role in boosting the overall effectiveness of the model, demonstrating its value in the context of complex multimodal data processing and retrieval tasks.

5) Bidirectional Distribution Matching: To ascertain the efficacy of our BDM loss, we conducted comparisons with the widely employed cross-modal projection matching (CMPM) loss [70] on two public datasets, as detailed in the comparison between No.1 and No.2. The results indicate that the BDM loss outshines the CMPM loss in R@1 accuracy, exhibiting substantial improvements of 4.56% and 5.38% on the respective datasets. Moreover, when contrasting the original InfoNCE loss with the CMPM loss (No.0 versus No.1), it was observed that the CMPM loss did not improve the retrieval performance for cross-modal RS images. In fact, there was a noticeable decline in performance. Conversely, implementing the BDM loss in place of the original loss led to an increase in R@1 accuracy by 1.05% and 1.08% on the two datasets. These findings strongly indicate that the BDM loss is more adept at adjusting feature representations between the two modalities. This superiority of the BDM loss plays a critical role in facilitating more precise and effective cross-modal interactions, thus validating its implementation in the model.

6) Multihead Cross Attention: As stated in Section IV-C, the introduction of a cross-attention mechanism in our MII has resulted in an increased number of parameters. Unlike the baseline, which lacks such cross-modal interaction mechanisms, we aimed to ascertain the significance of this additional parameter count. In Table VI, we compared our approach with two other popular methods of multimodal interaction. The multimodal interaction in MII is an efficient computational operation that fuses features from multiple modalities, facilitating the establishment of connections between them. Utilizing the same settings as our MII model, we conducted extensive comparisons with self-attention and merged attention. Our findings revealed slight but consistent improvements in performance on the rank metric with these methods. Remarkably, the primary advantage of our method lies in computational efficiency.

E. Zero-Shot CMRSITR

To validate the transfer capabilities of VLP-based methods and the MIIA framework, we trained different approaches on the RSICD dataset and subsequently tested them on two additional unseen datasets. All evaluation settings were consistent with those previously described. As shown in Table VII, it is evident that MIIA generally outperforms other methods across most metrics on both datasets, thus indicating robust transfer capabilities. Specifically, in the RSITMD dataset, MIIA achieves the highest score in text-to-image retrieval at Rank@1 with a score of 14.82% and achieves moderate image-to-text retrieval at Rank@1 with a score of 12.61%.

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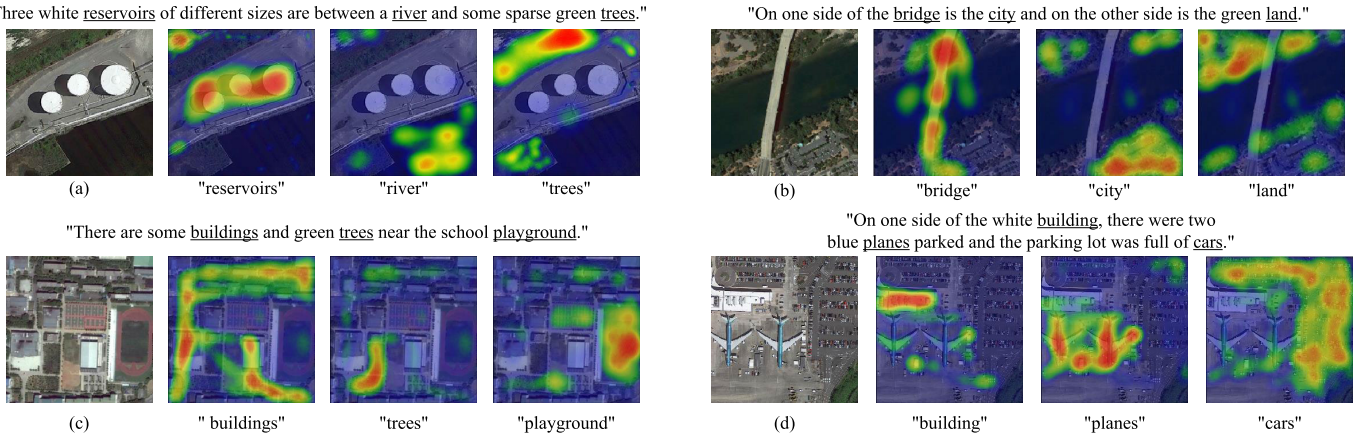


Fig. 9. Visualizations of attention maps of our MIIA using GAE [71] on image frames responding to individual words. (a)–(d) Image attention maps with respect to different words.

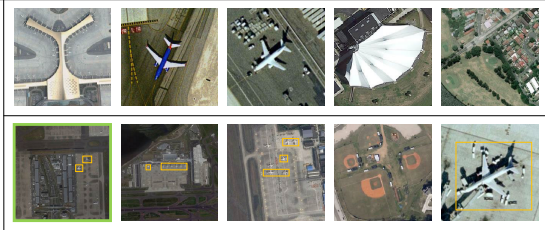
Query	Top-5 Retrieval Result	Query	Top-5 Retrieval Result
some white planes in the airport while with some dark buildings besides .			Some boats are located in the harbor surrounded by farmland and camps. Here are some Mooring Ships and ponygon bylands and ports Several boats are in a Port Surrounded by farmland and barley land. rows of ships docking in this port with only a narrow ship lock connected to the sea . The port was full of ships and several cars were parked on the shore. Some boats are located in the harbor surrounded by farmland and camps. Several ships were parked in a harbor surrounded by farmland and bare land. several boats are in a port surrounded by agricultural land and bare land. several rows of ships are docking in this port whose bank has buildings with red roofs . The ship is in the fields and wasteland around the dock
Near the blue factory, there are two parking areas with a grassy ground and a row of van near the road.			There are some buildings and three basketball courts near the school. There are many buildings and some green trees in school with two playgrounds. Many buildings and some green trees are in a school with two playgrounds. There are many buildings and green trees in the school, there are two baseball fields. There are some buildings and an amusement park with three basketball fields. Some buildings and a playground with three basketball courts are in a school. There are some buildings and three basketball courts near the school. Some buildings and green trees are at school in playground in three basketball fields. There are many buildings and some green trees in school with two playgrounds. Some buildings and some green trees are at school with playground and eight tennis courts.

Fig. 10. Comparison of the top-5 retrieval results for each query on the RSITMD dataset is presented. In each image-to-text or text-to-image query, the first row displays the baseline query results, while the second row shows the results from our MIIA method. Matched and unmatched images (or texts) are marked in green and black, respectively. The orange border represents the corresponding areas focused on by the MII module.

In addition, it shows competitive mR values, thereby further confirming its effectiveness. In the UCM dataset, although MIIA does not lead in all metrics, it maintains strong performance, particularly in I2TR@1 with a score of 12.11%, nearly reaching the highest score of 12.37% achieved by another method. It also demonstrates substantial mR values, indicating that it provides consistent and reliable retrieval capabilities across different datasets. Overall, these results underscore the potential of the MIIA in leveraging learned representations from the RSICD dataset to effectively handle different, unseen datasets, showcasing its adaptability and the effectiveness of its underlying mechanisms in VLP tasks. This capability is critical for practical applications where models must perform well across diverse data environments.

F. Qualitative Results

Fig. 9 displays the visual attention maps of our MIIA, highlighting its response to individual words on images. It is evident that the MII module adeptly locates different objects within the same image, even those of fine granularity. For instance, in response to "reservoirs," "rivers," and "trees," the attention is focused on the corresponding areas, demonstrating the model's high sensitivity to geographic features. Similarly, when responding to "buildings," "planes," and "cars," the model successfully differentiates between buildings, aircraft, and vehicles, even in cases where they are spatially close. These results indicate that the MII module is capable of effectively segmenting and identifying multiple objects in a scene, a crucial aspect of complex image understanding tasks. However, as shown in Fig. 9(d), our MIIA tends to focus on groups of cars, struggling with smaller objects or those overlapping in the image. This suggests that the model may require further optimization to address issues of object occlusion in complex scenes.

Fig. 10 depicts a side-by-side comparison of the top-5 retrieval results between the baseline model and our proposed MIIA framework. As demonstrated in the figure, MIIA consistently yields more accurate retrieval outcomes, particularly in instances where the baseline approach falls short. This enhanced accuracy can primarily be credited to the

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carefully designed MII module in our framework. The MII module is adept at leveraging fine-grained, distinctive features, a capability that is crucial in differentiating among various RS images. This effectiveness is particularly emphasized in Fig. 10, where specific text and image regions are marked in orange to demonstrate the improved alignment and correlation.

An interesting observation from our analysis is that while our model effectively captures word-level semantic information, it exhibits limitations in comprehensively understanding phrase-level semantics within descriptive texts. This can occasionally lead to semantic inaccuracies. A plausible explanation for this phenomenon could be the method employed in our MVLM implementation, where we masked random individual tokens rather than phrase-level segments. Identifying and resolving this limitation constitutes a key aspect of our future research endeavors, aiming to further enhance the model's comprehension of more complex, phrase-level semantic structures.

V. Conclusion

In this study, we introduce the MIIA framework, representing a novel and robust approach tailored for extracting distinctive image–text features in the RS domain. MIIA employs MVLM to thoroughly explore the fine-grained interconnections between visual elements and textual tokens. Complementing this, we have incorporated a cross-modal DCL approach and a BDM loss. These components have been meticulously designed to enhance the differentiation of nonmatching pairs and to reinforce the correlation among matching pairs, effectively aligning image–text pairs within a cohesive embedding space. The substantial enhancements observed on three widely utilized benchmark datasets demonstrate the superiority and effectiveness of the MIIA framework.

Despite its effectiveness, our current approach primarily focuses on word-level information in textual analysis, tending to overlook the more complex phrase-level semantics. This aspect is especially significant given the repetitive nature of text descriptions in public datasets. In our forthcoming research endeavors, we intend to address this limitation, with a focus on enhancing phrase-level semantic understanding. We foresee that VLP-based Methods will gradually become pivotal in the field of CMRSITR, propelling advancements and setting new benchmarks in the domain.

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